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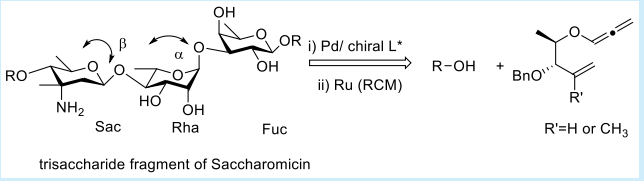
# Synthetic Study toward Saccharomicin Based upon Asymmetric Metal Catalysis

Bhawna Barpuzary, Mijin Kim, and Young Ho Rhee*

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**ABSTRACT:** Here, we report a *de novo* metal-catalyzed approach toward the stereoselective glycosidic bond formation in saccharomicin. The signature step is highlighted by the Pd-catalyzed asymmetric coupling of ene-alkoxyallenes and highly functionalized alcohol substrates. The reaction showed high chemo-, regio-, and ligand-driven diastereoselectivity. In combination with the ring-closing metathesis and late-stage functionalization, this method led to highly efficient synthesis of saccharosamine−rhamnose and rhamnose−fucose fragments.

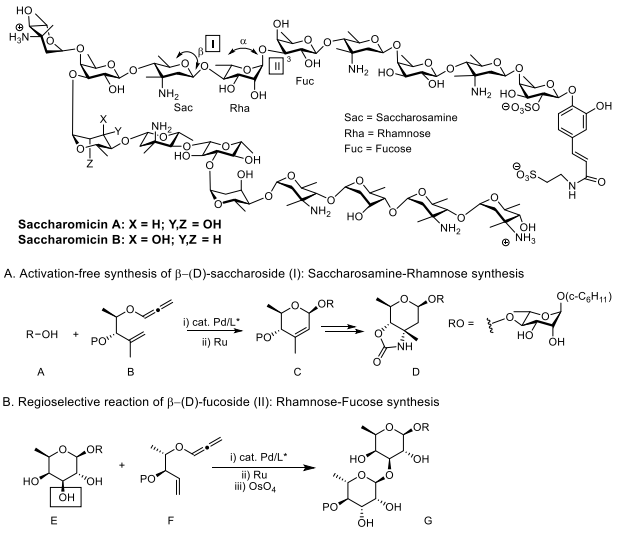



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Saccharomicins A and B were first isolated in 1998 from actinomycetes *Saccharothrix espanaensis*.[1] These natural products showed significant activities against various Gram-positive and Gram-negative bacteria as well as methicillin-resistant *Staphylococcus aureus* (MRSA). Thus, saccharomicins and their smaller structural analogues have the potential to be new antibacterial therapeutics. Chemical synthesis may play a significant role in providing candidates that help to explore structure−activity relationships of the saccharomicin and to delineate the mode of action.

From a structural viewpoint, saccharomicin A represents heptadecasaccharide linked to *N*-(*m*,*p*-dihydroxycinnamoyl) taurine aglycone. The oligosaccharide moiety in saccharomicin highlights the longest oligosaccharide found in any natural products. It consists of five deoxysugar monosaccharides including *β*-D-3-methyl-3-amino-2,6-dideoxysugars (saccharosamine), *β*-D-fucose, *α*-L-rhamnose, *α*-L-digitoxose, and *α*-L-3-epivancosamine. The daunting structure and interesting biological activities of saccharomicins have attracted significant attention from the synthetic community. A number of strategies have been developed to prepare for fragments of saccharomicin,[2−5] which mostly focused on the preparation of the core trisaccharide structure consisting of saccharosamine−rhamnose−fucose (Sac-Rha-Fuc in Scheme 1). We recently reported Pd-catalyzed asymmetric hydroalkoxylation of alkoxyallene, which successfully evolved into the unique synthesis of numerous 2-deoxyoligosaccharides as well as rare sugars with no need of activating groups.[6] Here, we report that glycosidic bonds abundant in saccharomicin (I and II in Scheme 1) could be constructed by this *de novo* method in a highly efficient manner. For example, Pd-catalyzed coupling reaction of readily available homoallylic alcohol A and ene-alkoxyallene B produced dihydropyran intermediate C, wherein the thermodynamically less stable *β*-stereochemistry was driven by the chiral ligand. Variation of the alkoxyallene structure in combination with the oxidative cyclization of carbamate derived from C delivered the disaccharide form of saccharosamine D linked to *α*-L-rhamnoside. Furthermore, analogous Pd-catalyzed coupling of unprotected *β*-D-fucose E and ene-alkoxyallene F built up the disaccharide G with high stereo- and regioselectivity.

**Scheme 1. Structure of Saccharomicins and the Key Concept of the Study**




At the outset of the study, we explored the synthesis of compound **7a**, a key precursor to *β*-D-saccharosamine (Table 1). We first envisioned that the intermediate acyclic acetal **4a** can be

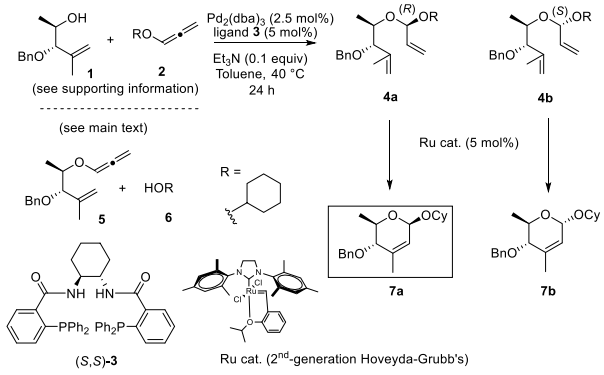
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**Table 1. Optimization of Synthesis of 7a**




| entry | ligand | equiv 5/6 | 1st step product (yield[a]) | 2nd step product (yield[b], ratio[c]) |
|---|---|---|---|---|
| 1 | (*S*,*S*)-**3** | 1:1 | **4a** (36%) | – |
| 2 | (*S*,*S*)-**3** | 2:1 | **4a** (48%) | – |
| 3 | (*S*,*S*)-**3** | 1:2 | **4a** (59%) | – |
| 4[d,e] | (*S*,*S*)-**3** | 1:2 | **4a** (65%[g]) | **7a** (52%, 8:1) |
| 5[f] | (*R*,*R*)-**3** | 1:2 | **4b** (94%[g]) | **7b** (90%, <1:25) |

[a]Determined by the integration of the crude $^1$H NMR. [b]Isolated yield of the pure major stereoisomer over two steps. [c]Ratio of **7a**:**7b** determined by the integration of the crude $^1$H NMR. [d]5 mol % of $Pd_2(dba)_3$ and 10 mol % of ligand were employed. [e]The reaction was run on a 1 mmol scale. [f]3.5 mol % of $Pd_2(dba)_3$ and 7 mol % of ligand were employed. [g]Isolated yield.

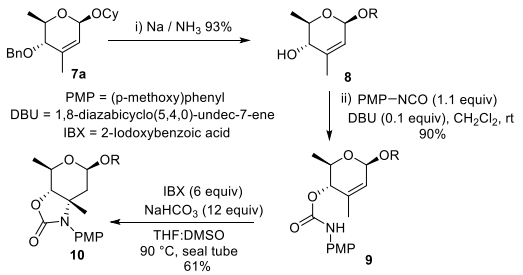
accessed by the Pd-catalyzed asymmetric coupling of readily available alcohol **1**[7] and alkoxyallene **2** in the presence of standard Trost-type ligand **3**. However, this reaction generated **4a** in only low (<20%) yield (for the detailed information, see the Supporting Information). Thus, we switched to an alternative method combining ene-alkoxyallene **5** and alcohol **6**. When an equimolar amount of these compounds was reacted in toluene at 40 °C for 24 h in the presence of $Pd_2(dba)_3$ (2.5 mol %), (*S*,*S*)-**3** (5 mol %), and a small amount of triethylamine, **4a** was obtained in significantly higher 36% yield (entry 1). Employing excess allene **5** (2 equiv) generated **4a** in higher yield (entry 2). Further improvement of the yield was seen with the use of excess alcohol **6** (entry 3). Finally, increasing the loading of the precatalyst $Pd_2(dba)_3$ (5 mol %) generated **4a** in 65% yield (entry 4). In this case, the subsequent RCM reaction using 5 mol % of second-generation Hoveyda–Grubbs catalyst provided the desired product **7a** in 52% yield over two steps with 8:1 stereoselectivity. The use of enantiomeric ligand (*R*,*R*)-**3** inverted the stereochemical outcome, generating **7b** in 90% yield with <1:25 stereoselectivity (entry 5). This result is consistent with our previous reports on various hydroalkoxylation reactions. Thus, the chiral-ligand-driven stereocontrol at the anomeric center worked well with sterically bulky allene **5**. The structure of **7a** was rigorously confirmed by various NMR techniques (for the detailed analysis, see the Supporting Information).

The next task was to synthesize **10** possessing all the stereochemical features of β-D-saccharosamine from the intermediate **7a** using the oxidative cyclization by way of the carbamate intermediate.[2a,8] For this purpose, the benzyl group in **7a** was first removed by $Na/NH_3$ to give the alcohol **8** in 93% yield. From this compound, carbamate **9** was uneventfully prepared in 90% yield by the reaction with *p*-methoxyphenyl isocyanate and DBU in $CH_2Cl_2$ at rt. The oxidative cyclization of **9** was conducted by heating with 2-iodoxybenzoic acid (IBX) and $NaHCO_3$ in DMSO:THF (10:1) at 90 °C, which produced the target **10** in 61% yield as a single diastereomer. Thus, the stereochemical purity of the β-glycosidic bond was not eroded in this oxidative cyclization.

Having established a protocol for the monosaccharide form of saccharomicin, we then expanded the reaction to the synthesis of saccharosamine-rhamnose disaccharide **17** (I in Scheme 1). For this purpose, allene **5** was reacted with readily available allylic alcohol **11** (2 equiv) possessing the stereochemistry found in the natural product. As depicted in Scheme 3, the asymmetric Pd-catalyzed coupling reaction turned out to be considerably faster than that of cyclohexanol **6** described in Table 1. In this case, the acyclic *O*,*O*-acetal **12a** was formed in near-quantitative isolated yield in the presence of 3.5 mol % of $Pd_2(dba)_3$ and 7 mol % of ligand (*S*,*S*)-**3** as a single stereoisomer.[9] This compound was then treated with second-generation Hoveyda–Grubb's catalyst (5 mol %) to provide the biscyclic intermediate **13a** in 79% yield with no erosion of the stereochemical purity. Notably, potentially competing ring-rearrangement metathesis (RRM) was not observed in this reaction.[10] Then, we employed the enantiomeric (*R*,*R*)-**3** ligand to determine the chiral ligand-driven stereoselectivity in the Pd-catalyzed reaction. In this case, diastereomeric **13b** was obtained again as a single stereoisomer by way of **12b** in 86% yield over two steps. The NMR spectrum of **13a** was clearly distinguishable from that of **13b**, thus confirming the stereoselectivity described above (for the details of the analysis, see the Supporting Information).

Synthesis of the disaccharide **17** requires late-stage chemoselective dihydroxylation that can distinguish two olefin moieties in **13a**. We reasoned that this selectivity could be easily induced by the contrasting stereochemical feature of anomeric centers. Indeed, subjection of the compound **13a** to the Upjohn dihydroxylation condition[11] using $OsO_4$ (3 mol %) and *N*-methylmorpholine oxide (1.3 equiv) generated diol **14** in 81% yield as a single compound with complete chemo- and stereoselectivity. The subsequent treatment of diol **14** by methoxymethyl chloride (MOMCl) provided **15** in 89% yield. Synthesis of fully functionalized disaccharide **17** from **15** proceeded uneventfully using the protocol developed in Scheme 2. Thus, removal of the benzyl group using $Na/NH_3$ (93% yield)

**Scheme 2. Synthesis of 10 from 7a by the Oxidative Cyclization**

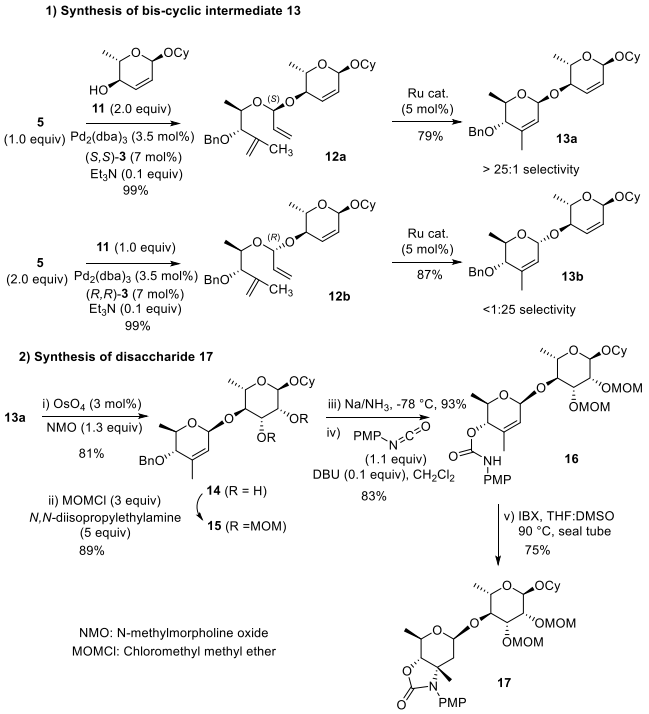



followed by treatment with DBU and *p*-methoxyphenyl isocyanate produced carbamate **16** in 83% yield. This compound was heated with IBX in DMSO:THF (10:1) in the presence of $NaHCO_3$, in a sealed tube at 90 °C to give **17** in 75% yield (Scheme 3).

Having prepared the saccharosamine–rhamnose disaccharide moiety, our final efforts were directed at the construction of the glycosidic-bond-connecting fucose–rhamnose disaccharide

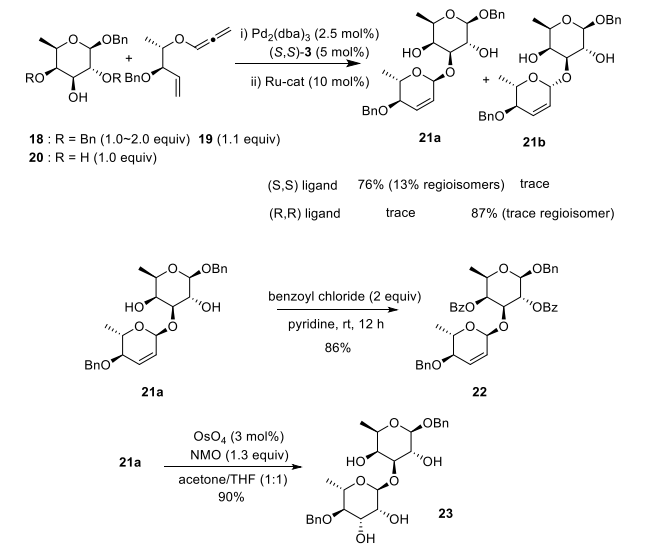

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**Scheme 3. Synthesis of the 3-Amino-3-methyl Disaccharide Unit of Saccharomicin**




moiety (II in Scheme 1). This reaction necessitates selective coupling reaction at $C_3$−OH of $\beta$-D-fucoside. Preliminary tests using the substrate **18** equipped with protective groups at other hydroxyl groups and the allene **19**[6d] showed the formation of an adduct in only trace yield, presumably due to the steric congestion of the substrate **18** (Scheme 4). Remarkably, the use of unprotected triol **20** significantly improved the result. For example, the use of **20** (1 equiv) and slight excess of **19** (1.1 equiv) in the presence of $Pd_2(dba)_3$ (2.5 mol %), ligand (*S*,*S*)-**3** (5 mol %), and triethylamine (0.1 equiv) generated the desired compound **21a** in 76% yield after RCM reaction along with a small amount of regioisomers (~13% combined yield). In this case, the stereoisomer **21b** was obtained only in trace amount. Employing enantiomeric ligand (*R*,*R*)-**3** provided **21b** in 87% yield with the formation of regio- and stereoisomers in a trace amount.[12] The regiochemical outcome of the reaction was confirmed by the extensive NMR studies of bisbenzoate **22**, which could be readily prepared from **21**. (For the detailed information on the synthesis and analysis of **22**, see the Supporting Information.) The final dihydroxylation reaction went smoothly to provide the tetrol **23** in 90% yield as a single diastereomer.

**Scheme 4. Synthesis of Fucose−Rhamnose Disaccharide by Protective Groupless Coupling**




In summary, we have developed a *de novo* synthetic method toward saccharomicin representing a formidable challenge in deoxyoligosaccharide synthesis. The unique regio-, chemo-, and stereoselectivities address the utility of various metal-catalyzed reactions in this endeavor. Currently, we are working on the development of convergent synthetic strategies aiming at the total synthesis of saccharomicin. The results of this study will be reported in due course.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02060.

Experimental data and copies of $^1H$ and $^{13}C$ spectra for all new compounds (PDF)

FAIR data, including the primary NMR FID files, for compounds **1**, **4**, **5**, **7**−**10**, **12**−**17**, and **21**−**23** (ZIP)

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### Notes

The authors declare no competing financial interest.

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